PES UNIVERSITY (ECC)

DEPARTMENT OF CSE

SOFTWARE ENGINEERING — LAB 3

Component Modelling & Architectural Pattern Selection

Problem Statement #20 — Hospital Bed & ICU Allocation Dashboard

Abhirama H Y • PES2UG24CS023 • Section A

ARCHITECTURE SELECTION

- **I chose Microservices Architecture for the Hospital Bed & ICU Allocation Dashboard.**

- **Architectural Choice:** Microservices Architecture is selected because the dashboard contains distinct operational capabilities for triage and ER bed allocation, real-time bed status, ICU transfer prioritization, inter-facility ambulance transfers, critical-event notifications, and role-based access control.

- **Reason 1:** The system can scale and isolate faults by workflow. High-demand functions such as real-time bed status and triage can be scaled independently, while failures in one transfer or notification service need not stop unrelated hospital workflows.

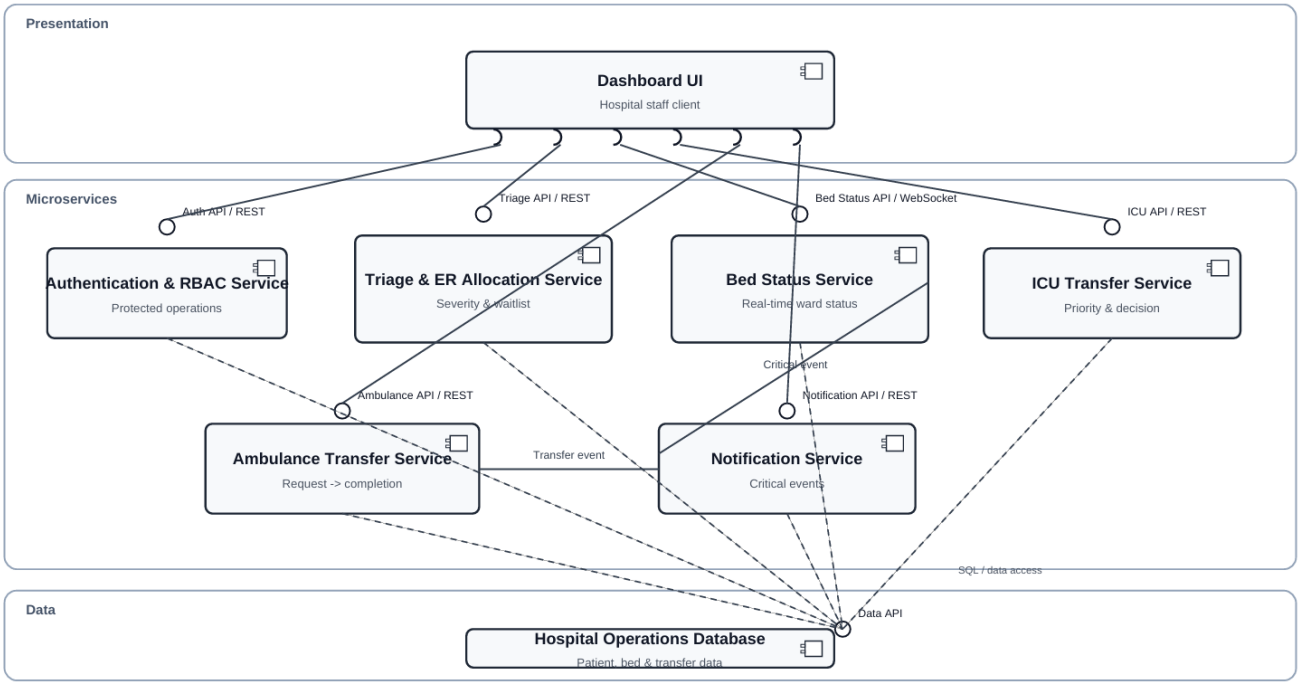
- **Reason 2:** Each major workflow can be implemented and deployed as an independently managed service with clear interfaces. This supports the separate ER, ICU, ambulance-transfer, and notification functions defined in the requirements.

- **Security Advantage:** A dedicated Authentication & RBAC Service can enforce authenticated, role-authorized access to protected triage, bed-allocation, administration, and ambulance-transfer actions, supporting NFR-002.

- **Performance Benefit:** Independent service scaling and separation of workloads support responsive real-time operations. The Bed Status Service can handle WebSocket-based status propagation to connected dashboard clients, supporting the under-1-second target in NFR-001.

Hospital Bed & ICU Allocation Dashboard

UML Component Diagram - Microservices Architecture



Ball = provided interface Socket = required interface Solid = service communication Dashed = data-access dependency